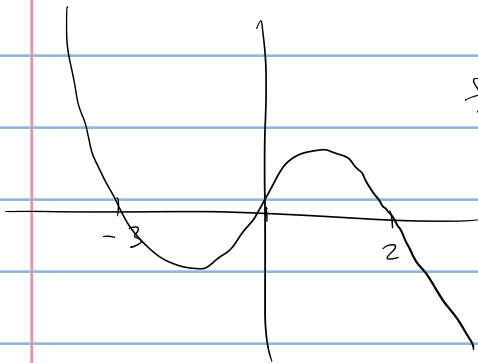


$f(x+a)$, $f(x)+a$, $f(ax)$, $af(x)$
Transformations

To find a polynomial with specific roots

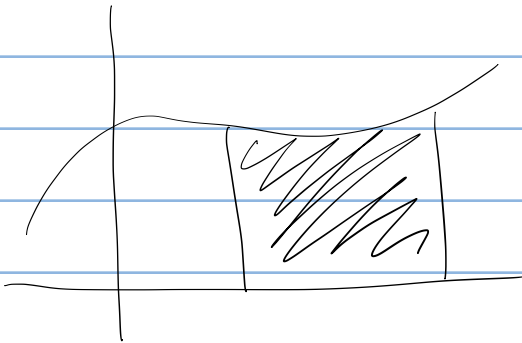
For example: Find a polynomial at $x = -3, 0, 2$



$$f(x) = -(x+3)x(x-2)$$

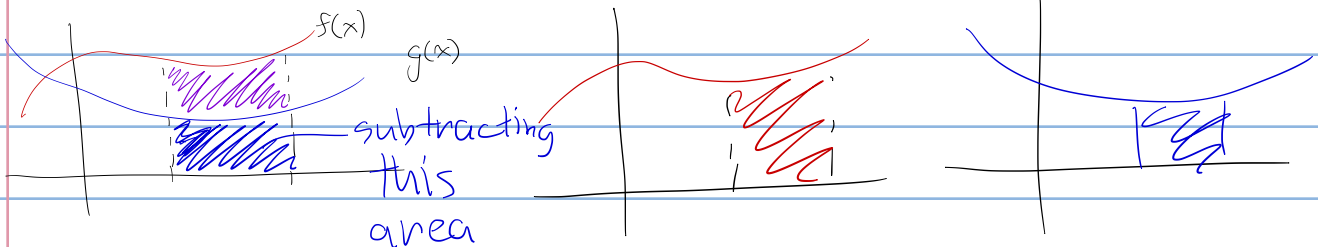
6.3. Applications of Integrals

Integral represents area under curve



6.3.1 Areas

Areas between curves



Area between $f(x)$ and $g(x)$ is $\int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b [f(x) - g(x)] dx$

6.3.2 Cumulative Change

Find distance traveled

$$\frac{ds}{dt} = 50 \text{ mph}$$

↑
velocity

$$\frac{ds}{dt} = 50$$

$$\int_0^t \frac{ds}{dt} dt = \int_0^t 50 dt$$

$$s(t) - s(0) = 50t \Big|_0^t$$

$$s(t) - s(0) = 50t$$

$$s(5) \quad \uparrow = 50(5) = 250 \text{ mi}$$

start
from 0

$$a = -9.8 \text{ m/s}^2 = -32 \text{ ft/s}^2$$

$$a = \frac{dv}{dt}$$

$$\int -9.8 dt = v(t) - v(0)$$

$$-9.8t = v(t) - v(0)$$

↑
initial
velocity

$$\int (v(t) = -9.8t + v_0) dt$$

$$s(t) - s(0) = -\frac{1}{2} \cdot 9.8t^2 + v_0 t$$

$$s(t) = -\frac{1}{2} \cdot 9.8t^2 + v_0 t + s_0$$

How long does a ball take to fall from the top of a building that is 10 m tall?

$$a = -9.8 \text{ m/s}^2$$

s_0 = height of the building

$v_0 = 0 \text{ m/s}$ since ball is dropped

$$s(t) = -\frac{1}{2} \cdot 9.8 \cdot t^2 + v_0 t + s_0$$

$$s(t) = 0 = -\frac{1}{2} \cdot 9.8 t^2 + 10$$

$$t \approx 1.43 \text{ s}$$

What is the population of a colony of bacteria if the growth rate is $0.5t$ where t is in hours and has an initial population of 500 after 7 hours

$N(t)$ is the population of the bacteria

$$\int_0^t \left(\frac{dN}{dt} = 0.5t \right) dt$$

$$\begin{aligned} N(t) - N(0) &= \int_0^t 0.5t \, dt \\ &= \frac{1}{2} 0.5 t^2 \Big|_0^t \end{aligned}$$

$$N(t) = 0.25 t^2 + N(0)$$

$$N(t) = 0.25 t^2 + 500$$

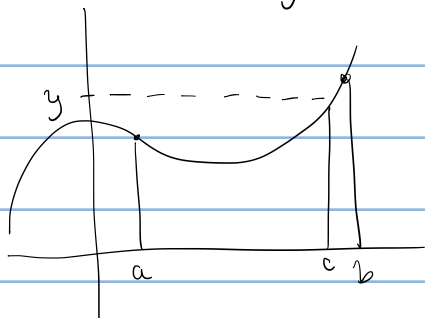
$$N(7) = 0.25(7)^2 + 500$$

$$= 12.25 + 500$$

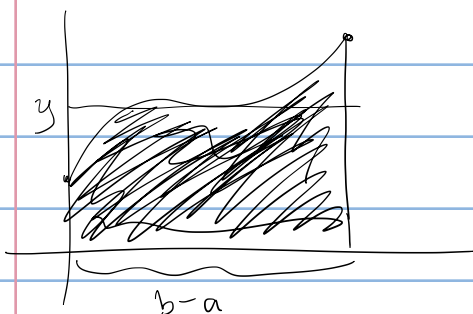
$$= 512.25$$

$$\approx 512$$

6.3.3. Average Values



If $f(x)$ is continuous and y is between $f(a)$ and $f(b)$, there exists a c between a and b such that $f(c) = y$.

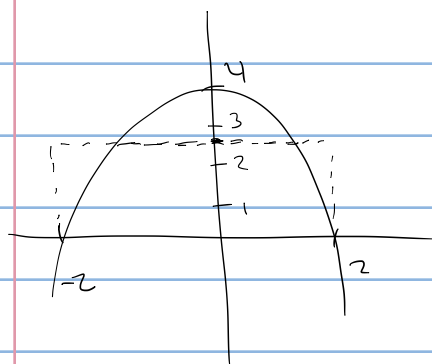


$$A = \int_a^b f(x) dx$$

$$y(b-a) = \int_a^b f(x) dx$$

$$y = \frac{1}{b-a} \int_a^b f(x) dx$$

Find the average value of $y = 4 - x^2$ on the interval $[-2, 2]$.

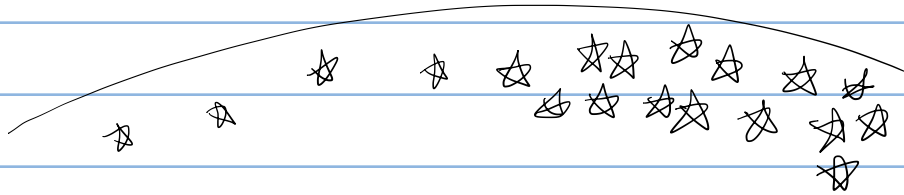


$$y = \frac{1}{2 - (-2)} \int_{-2}^2 (4 - x^2) dx$$

$$= \frac{1}{4} \left(4x - \frac{1}{3} x^3 \right) \Big|_{-2}^2$$

$$= \frac{1}{4} \left[8 - \frac{8}{3} + 8 - \frac{8}{3} \right] = \frac{8}{3}$$

Cumulative change example



The density of starfish along a beach is modeled by

$$y = 3 + 4x$$

Find the total amount of starfish along the beach if the beach is 5 miles long.

$$\int_0^x \frac{dN}{dx} = 3 + 4x \, dx$$

$$N(x) - N(0) = 3x + 2x^2 \Big|_0^x$$

$$N(x) = 3x + 2x^2$$

$$N(5) = 3 \cdot 5 + 2 \cdot 5^2$$

$$= 15 + 50$$

$$= 65 \text{ starfish}$$